

High-Frequency Modeling of Frequency-Dependent Dielectric and Conductor Losses in Transmission Lines

Jason Chan, Kondagunta Sivaprasad and Kent Chamberlin

Abstract— This paper details the implementation of three theoretical techniques to calculate dielectric and conductive losses in transmission lines at high frequencies. Those techniques are the Kramers-Kronig relationship to account for the frequency-dependent loss tangent, inclusion of surface roughness in estimating conductor losses, and a method for calculating moisture effects. When these mechanisms are incorporated into a contemporary stripline model, the high-frequency results are in much closer agreement with measured data than when they are not included.

Index Terms—Broadband, Frequency Dependent, Loss Tangent, Modeling, Stripline

I. INTRODUCTION

THE objective of the work described here is to apply existing theoretical approaches to better estimate high-frequency losses in transmission lines. Those approaches individually account for the frequency-dependent losses in the dielectric, the effects of moisture content in the dielectric and surface roughness effects on the conductor. While these findings can be used to calculate the frequency-dependent losses for any transmission line structure, they are applied here to stripline transmission lines embedded in FR-4 (Fire Retardant #4). A stripline in FR-4 was chosen for this study because it is the most commonly used transmission line for the frequency regime addressed here and because high-quality measured loss data were available for it.

Comparisons of measured and modeled loss data using contemporary modeling approaches performed at the start of this effort revealed significant errors beginning at about 2 GHz, making those approaches unsuitable for modeling applications such as long backplane interconnects at high frequencies. This paper identifies the major mechanisms affecting stripline losses, and shows how they can be implemented in a model. The close agreement between the final model results and measured data demonstrates that those loss mechanisms are

accounted for correctly.

The first facet of this task was to collect high-fidelity, wide-spectrum loss data for strip lines embedded in FR-4. Loss data for the striplines were collected for frequencies up to 14 GHz. Those measurements are described in the following section along with a comparison of modeled loss data calculated using a contemporary approach. That approach assumes a constant loss tangent, independent of frequency, in calculating dielectric loss, along with a conduction loss model described by Collin [1]. It was the disagreement between measurements and the existing model in that initial study that led to the work presented here.

Measurements of the loss tangent for good dielectrics with very low loss are often inaccurate. For this reason, and because it has been shown to provide reasonable results in some applications, a constant loss tangent value is generally assumed when modeling broad-band stripline performance. As is demonstrated by the measured and modeled data comparisons given in Section II, that assumption leads to increasing modeled errors at higher frequencies. As is shown below, these errors are ameliorated by applying the Kramers-Kronig [2] relations to estimate the dispersive loss tangent using the more accurately measured dielectric constant of FR-4.

The dielectric properties of the FR-4 used in this study were taken from industrial datasheets for dry samples [3,4], although the loss measurements presented in this paper were taken after the stripline had been exposed to moisture for a long period of time. Because the dielectric constant was determined using a dry sample, the effects of moisture had to be incorporated into the Kramers-Kronig relations for this study. Although this effect does not need to be accounted for in cases where the dielectric properties are measured using a sample with a standardized moisture content, it does demonstrate explicitly the effects of moisture on stripline losses. The effects of moisture, and the approach used to model the dispersive loss tangent are given in Section II.

The conduction loss per unit length is typically determined using a conduction loss model [1] that assumes smooth conductors, although it is recognized that conductor roughness can also affect loss [5,6]. An existing technique to account for conductor roughness and the application of that technique [7] to a contemporary stripline loss model is also described.

When operating at frequencies below higher-order mode cutoff, it is reasonable to assume TEM wave propagation. For the stripline geometry used in this study, that cutoff is around 10 GHz [8]. The model results presented here were made

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assuming TEM propagation, and it is shown below that the final model results are in close agreement even well above that cutoff. However, it is recommended that the results presented here be applied only for frequencies below higher-order mode cutoff, which for the configuration being evaluated here is 10 GHz. as stated above.

The final result of the work presented here is a model that accounts for the above loss mechanisms. This model allows for a more accurate prediction of stripline loss than has been reported previously, as is demonstrated by the close agreement between measured and modeled data.

II. STRIPLINE LOSS MEASUREMENTS AND COMPARISON WITH CONTEMPORARY MODEL DATA

A. Contemporary Model for Estimating Stripline Losses

A contemporary stripline transmission line model uses a conformal mapping solution given by Collin [1], as is summarized below. The model assumes a symmetric stripline as shown in Figure 1 immersed in a dielectric ($\epsilon_r(\omega)$). The center conductor is of width, w , and thickness, t , and is separated from the outer conductors by a distance, d .

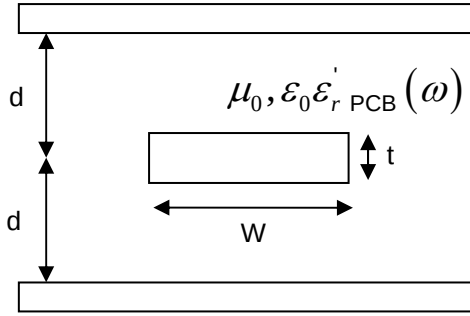


Figure 1: Cross-Sectional Parameters of a Stripline

The total dispersive attenuation per-unit-length, due to the conduction and dielectric losses, is

$$\alpha_{Total}(\omega) = \alpha_{center}(\omega) + \alpha_{ground}(\omega) + \alpha_{dielectric}(\omega) \quad (1)$$

where:

$$\alpha_{center}(\omega) = \frac{\pi R_m}{16 Z_c d (K')^2 k'} \ln \left[\frac{16 d k'}{k \pi T_e} \right] \quad (2)$$

$$\alpha_{ground}(\omega) = \frac{\pi^2 R_m W}{64 Z_c d^2 [K(k')]^2 k'} \quad (3)$$

$$\alpha_{dielectric}(\omega) \approx \frac{\omega \sqrt{\mu_0 \epsilon_0}}{2} \tan(\delta) \quad (4)$$

where R_m is the surface resistance of copper [9]. Additionally, the loss tangent is assumed to be constant. The characteristic impedance of the stripline, Z_c is given by:

$$Z_c = \frac{\sqrt{\mu_0 \epsilon_0 \epsilon'_r_{PCB}(\omega)}}{C} \quad (5)$$

where the dispersive dielectric constant of the PCB, $\epsilon'_r_{PCB}(\omega)$, is known. The capacitance per unit length, C , is given by:

$$C = 2 \epsilon_0 \epsilon'_r_{PCB}(\omega) \frac{K(k')}{K(k)} \quad (6)$$

$K(k)$ and $K(k')$ are elliptic integrals of the first kind, where the parameters k and k' are given by:

$$k = \frac{1}{\cosh \left(\frac{\pi W}{4d} \right)} \quad (7)$$

$$k' = \tanh \left(\frac{\pi W}{4d} \right)$$

The equivalent thickness, T_e , is determined by:

$$T_e = e^{-\frac{\pi}{2} \sqrt{\frac{4Wt}{\pi}}} \quad (8)$$

B. Comparison of Measured losses vs. Contemporary Model Losses

An FR-4 backplane test board, supplied by Teradyne Inc., had stripline lengths of 10 and 20 inches. These striplines had a center conductor thickness of $t = 1.2$ mil and width of $w = 12$ mil. As the supplied striplines were slightly asymmetric, an average symmetric distance of $d = 12.4$ mil was used. The S_{21} scattering parameters of these striplines were measured with a Hewlett Packard 8722D Vector Network Analyzer. The dispersive attenuation per-unit-length of each stripline length was determined from their respective S_{21} measurements, and the measured results for both lines agreed closely with each other. A plot of the measured dispersive attenuation per-unit-length is shown in Figure 2; the anomaly seen in the plot from around 9 to 11 GHz is likely caused by launch resonance. The NIST-proposed, multi-line calibration method to de-embed the SMA launches from the measured striplines to extract the frequency-dependent, complex propagation constant [10] was investigated. The de-embedding process has been shown to be applicable to striplines [11], although little difference was seen at frequencies up to 14 GHz when applying the de-embedding method to the data collected for this study. Because of this, no attempts were made to remove the effects of the connectors and vias in the measured data presented in this paper.

As is discussed in greater detail in Section III B, the loss tangent for FR-4 reported by the manufacturer is approximately 0.015. Consistent with the contemporary modeling approach, this value was assumed to be independent of frequency. Further, it was assumed that the published loss tangent accounted for the effects of typical moisture. The loss-tangent value along with the stripline dimensions were used in the contemporary model to estimate loss as a function of frequency, and that loss is plotted in Figure 2 along with the measured values. As seen in Figure 2, the contemporary model is in reasonable agreement with measured coefficients for frequencies below 2 GHz, although model values tend to diverge from measured losses at higher frequencies. This observation suggested that additional loss mechanisms in the dielectric and conduction-loss estimation techniques needed to be included in order to predict stripline losses more accurately.

The measured and modeled results plotted in Figure 2 are fundamental to this paper since they reveal the significant error associated with a commonly-used modeling approach. Because of this, the data plotted in Figure 2 are used as benchmarks in all of the model comparison data shown below.

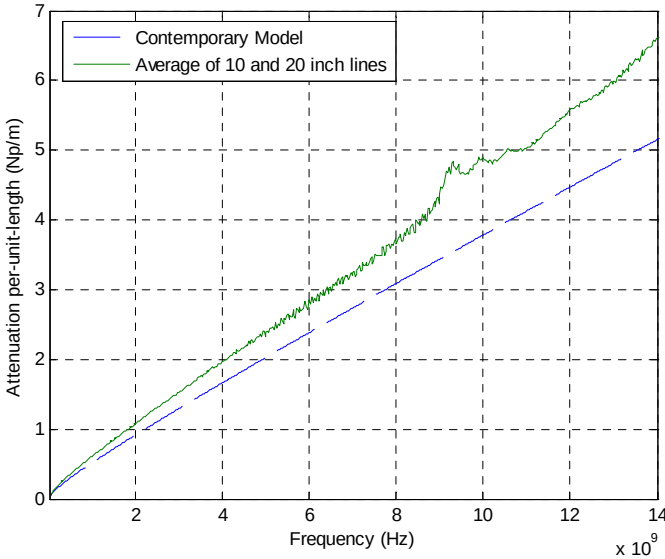


Figure 2: Contemporary Model vs. Measured Attenuation Coefficients

III. DETERMINING A MORE ACCURATE DISPERSIVE LOSS TANGENT

The contemporary model does not take frequency-dependent permittivity or conductor roughness into account when making loss estimates. The objective of this section is to outline the approaches that were used to model those mechanisms, with the following section showing how those mechanisms affect overall loss estimates compared against measured data.

While there are a number of different meanings for the term dispersive in the context of transmission-line losses, it is used here as an indication that the loss tangent is a function of fre-

quency. Hence, the contemporary model is a non-dispersive model, and the model described here is dispersive.

This section begins with an approach for modeling moisture effects, followed by the application of the Kramers-Kronig relationship and concluding with conductor roughness effects.

A. Modeling the Effects of Moisture

As stated in the introduction, measured permittivity data were available only for dry FR-4 samples, whereas the FR-4 used to collect loss data were not dry. Consequently, the effects of moisture had to be accounted for, and a technique is described in this section based upon previously referenced work. While rigorous validation would be prudent before using this approach on a broad scale, its implementation here is used to demonstrate the relationship between moisture content and loss. It is reassuring to note that the results obtained using realistic values of moisture content are in close agreement with measured data.

A mixture model was used to approximate the effect of the moisture embedded in FR-4. A mixed dielectric phenomenological model, assuming sparse prolate ellipsoidal moisture particles, is given by [12]:

$$\epsilon_{r_{eff}}'(\omega) = \epsilon_{r_{PCB}}'(\omega) + \frac{1}{3}v \frac{(\epsilon_{r_{moisture}}'(\omega) - \epsilon_{r_{PCB}}'(\omega))(\epsilon_{r_{moisture}}'(\omega) + 5\epsilon_{r_{PCB}}'(\omega))}{(\epsilon_{r_{moisture}}'(\omega) + 5\epsilon_{r_{PCB}}'(\omega))} \quad (9)$$

where v is the fractional volume of moisture, $\epsilon_{r_{moisture}}'(\omega)$ is the frequency-dependent permittivity of moisture, and $\epsilon_{r_{PCB}}'(\omega)$ is the dispersive measured permittivity of a dry FR-4 laminate. It should be noted that the accuracy of this effective dielectric model is not precisely known, since no references or past experiments have been found to prove the direct viability of the mixture model in the GHz range for varying ranges of moisture content. However, indications are that it should provide a reasonable estimate of the effective dielectric constant of FR-4 with sparse moisture content.

The measured frequency-dependent dielectric constant of dry FR-4 with 50% resin content is given in a non-published data sheet provided by the manufacturer (Pocyclad). As specified in the data sheet, the manufacturer obtained these measurements using two separate industry-standard measurement methods. The low frequency values (1 MHz – 1.5 GHz) were obtained with the parallel plate resonator technique [13] while the remaining high frequency values (1.5 GHz – 20 GHz) were obtained with the stripline resonator technique [14,15]. The data were extrapolated beyond the actual measurement range (14 GHz – 20 GHz) using the approach prescribed by [2]. The data sheet also specifies that the loss tangent is approximately 0.015, which is assumed to be constant for all frequencies when using the contemporary model. Relative permittivity values are plotted in Figure 3 for both dry FR-4 and FR-4 with moisture content, calculated using Equation (9).

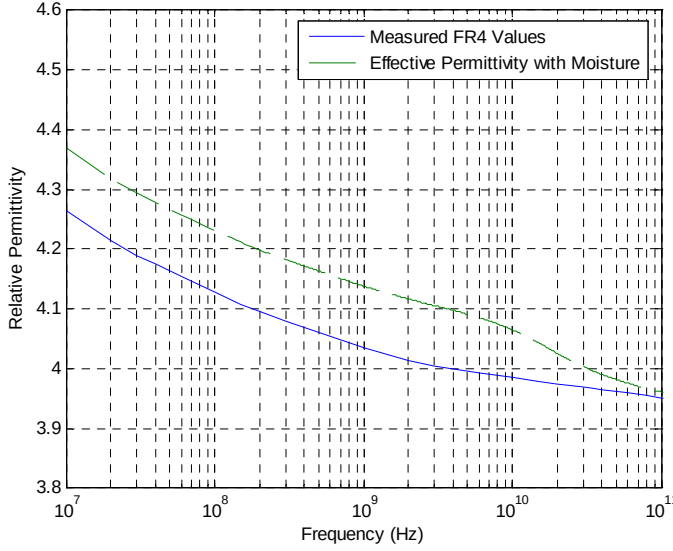


Figure 3: Real Component of FR-4 Relative Permittivity with and without Moisture

The measurements of stripline loss were conducted with an indoor temperature of approximately 20°C and a relative humidity of 23.5%. The dielectric properties of water at 20°C, supplied by King, were used to calculate the real permittivity component of moisture, $\epsilon_{r \text{ moisture}}'(\omega)$, from 1 MHz to 400 GHz [16].

There was no precise method for determining the fractional volume content of moisture embedded in the backplane sample used in this study. However, past experimental studies have shown that the moisture content of FR-4 for the conditions given above is roughly 0.00185 by weight [3]. Thus, the equivalent fractional volume content can be calculated by the product of the weight fraction and the specific gravity of FR-4. As the typical specific gravity of FR-4 is roughly 1.85, the equivalent fractional volume content is $v = 0.003426$. The effective dielectric constant was determined using v , $\epsilon_{r \text{ PCB}}'(\omega)$, and $\epsilon_{r \text{ moisture}}'(\omega)$, shown in Figure 3. The effective dielectric constant was then applied to Equation (10) below to extrapolate the imaginary component of the dielectric constant, $\epsilon_{r \text{ eff}}''(\omega)$. Lastly, the dispersive $\epsilon_{r \text{ eff}}'(\omega)$ and $\epsilon_{r \text{ eff}}''(\omega)$ were used in Equation (11) to estimate the dispersive attenuation coefficient of the FR-4 dielectric material.

B. Using the Kramers-Kronig Relationship to Account for the Frequency-Dependent Loss Tangent

Since the real part of the complex permittivity for low-loss dielectrics such as FR-4 is more accurately determined than the imaginary part, there are clear advantages to using the Kramers-Kronig Relationship to calculate that parameter. Specifically, Kramers-Kronig specifies an integral equation that relates the real to imaginary component of the complex dielectric constant in a linear and causal medium (a valid assumption for lossy PCB's) where [2]:

$$\epsilon_{r \text{ eff}}''(\omega) = -\frac{2\omega}{\pi} \int_0^\infty \frac{\epsilon_{r \text{ eff}}'(\omega') - \epsilon_{r \text{ eff}}'(\omega)}{\omega'^2 - \omega^2} d\omega' \quad (10)$$

Since the real and imaginary effective permittivity components are known, equation (4) can be modified as:

$$\alpha_{\text{dielectric}}(\omega) \approx \frac{\omega \sqrt{\mu_0 \epsilon_0}}{2} \left(\frac{\epsilon_{r \text{ eff}}''(\omega)}{\epsilon_{r \text{ eff}}'(\omega)} \right) \quad (11)$$

The values for $\epsilon_{r \text{ eff}}'(\omega)$ and $\epsilon_{r \text{ eff}}''(\omega)$ used in Equation (11) are derived in the previous section.

It should be noted that a multi-pole Debye Relaxation model [17] can also be used to estimate the imaginary part of the complex permittivity from the real part. That approach was investigated for the work presented here, although it was abandoned due to uncertainties about implementation, such as the number of Debye poles to use. Alternatively, the Kramers-Kronig Relationship was applied since the integral accounts for all possible poles without having to determine Debye parameters through numerical experimentation. Since information on both $\epsilon_{r \text{ eff}}'(\omega)$ and $\epsilon_{r \text{ eff}}''(\omega)$ are available after applying the Kramers-Kronig relationship of Equation (10), the multi-pole Debye model can be determined with certainty. This method for determining Debye parameters could prove instrumental in the Finite-Difference, Time-Domain (FDTD) modeling of frequency-dependent dielectrics, as the Debye approach is computationally more efficient in FDTD [18].

C. Accounting for the Effects of Conductor Roughness

The surface resistance in the contemporary model is calculated assuming perfectly-smooth copper planes. However, the sample board for the work presented here utilized 1 oz. electrodeposited copper to bond the center and ground conducting planes to the FR-4 laminate. Since documentation has shown that the surface roughness of electrodeposited copper can exacerbate insertion losses [5,6], the surface resistance of the contemporary model was modified to approximate the effect of the rough surface. Surface resistance can be modified by incorporating a correction coefficient as:

$$R_m = \frac{1}{\sigma_{Cu} \delta} \left(1 + \frac{2}{\pi} \tan^{-1} \left(1.4 \frac{d_{rms}}{\delta} \right)^2 \right) \quad (13)$$

where d_{rms} is the RMS surface roughness and δ is the skin depth of copper [7]. An RMS surface roughness height value of $d_{rms}=2.4 \mu\text{m}$ was used for the 1 oz. copper [6]. The exposed sides of the core and pre-preg layers were initially “smooth” prior to PCB fabrication. In order to bond the core and pre-preg layers, the exposed copper layers were treated to roughen the surfaces even further. The fabrication process data for the sample card indicated that Alpha-Prep treatment was applied, meaning that the degree and profile of roughness

is approximately equivalent to the “pre-roughen” copper foil layers.

A microscopic picture is given in Figure 4 to illustrate the magnitude of surface roughness relative to the size of a typical stripline structure.

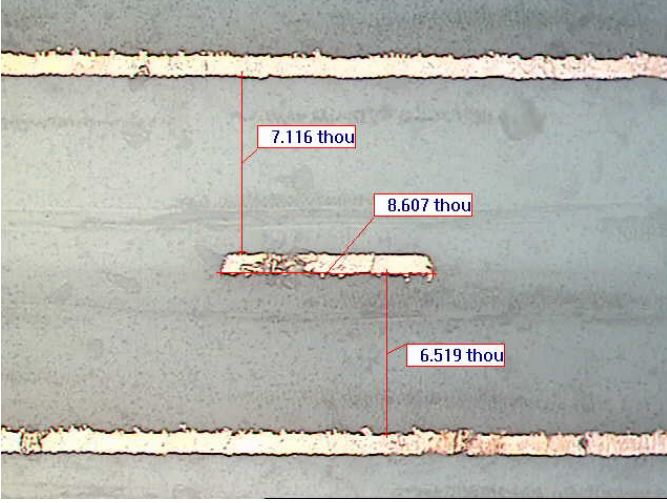


Figure 4: Cross Section of Typical Stripline to Show Conductor Roughness. The label “thou” Represents Thousandths of an Inch.

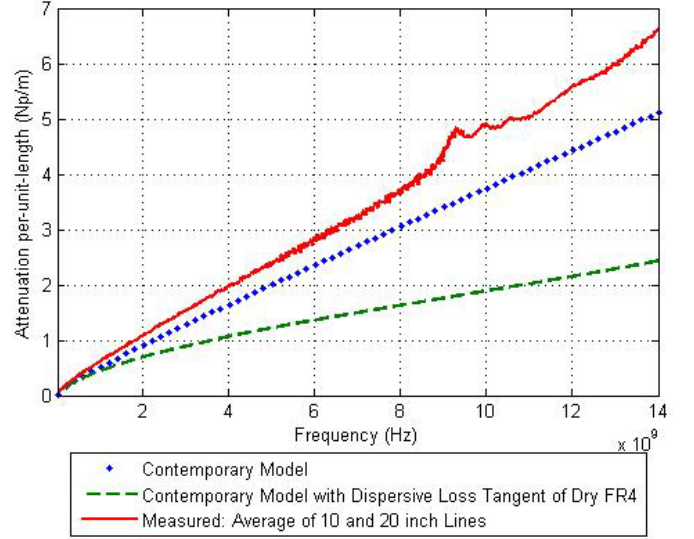
There are several caveats that should be mentioned here. First, it is recognized that Equation (13) is an approximation that applies only to square and triangular conductor profiles. And second, the roughness term, d_{rms} , can vary from vendor to vendor. Cognizant of these potential limitations in modeling rough conductors, the approach here is to document details of the modeling process along with the results obtained in an effort to facilitate future work in this arena should it be pursued. For those wishing to investigate the effects of roughness on other than square or triangular geometries, it is noted that the original work which subsequently led to Equation (13) can be found in [19].

IV. MEASURED AND MODELED DATA COMPARISONS

The objective of this section is to illustrate the effects of the loss mechanisms outlined in the previous sections and to show how the losses calculated when accounting for those mechanisms compare with measured data.

The first loss mechanism presented here is loss due to moisture in the dielectric, and it is plotted in Figure 5. As expected, modeled loss for the dry dielectric is less than the loss for the dielectric with moisture ($v = 0.003426$). For this plot, a frequency-dependent loss tangent is assumed, as are smooth conductors. The greater underestimation of loss using the dispersive loss tangent for dry FR-4 illustrates the dominant effect of moisture content on loss. In particular, the calculated dispersive loss tangent of the dry FR-4 material is lower than the presumed constant loss tangent of 0.015 used in the contemporary model. As is shown below, the agreement between measured and modeled data become much closer as the effects of moisture are included.

Figure 5: Contemporary Model with the Dispersive Loss Tangent of Dry FR-4 Plotted with Benchmark Measured and Modeled Data



For the remainder of the data comparisons below, the model was run using parameters for a dielectric containing moisture, because the measured data were collected on a stripline with a dielectric containing moisture. Figure 6 plots loss versus frequency for the dispersive loss tangent of FR-4 including moisture along with the benchmark data. The significant impact of moisture on loss is evident in Figures 5 and 6.

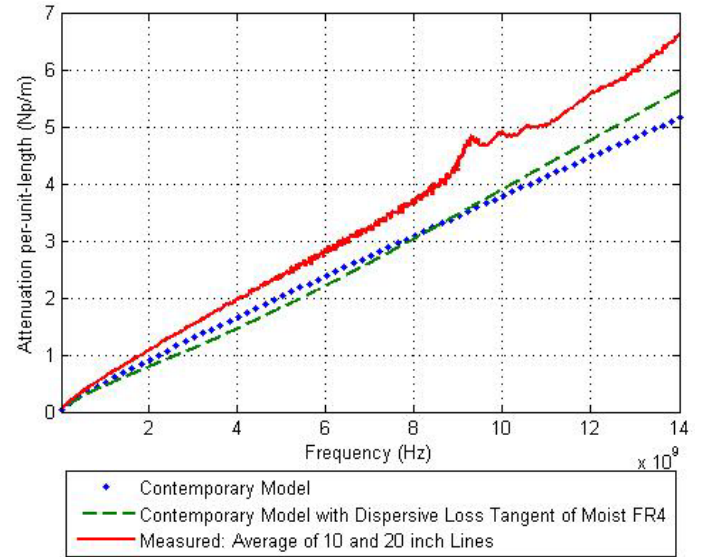


Figure 6: Contemporary Model with the Dispersive Loss Tangent of FR-4 including Moisture Plotted with Benchmark Measured and Modeled Data

Figure 7 plots loss when frequency-dependence and conductor roughness are incorporated into the model. Close agreement between measured and modeled results are evident for the frequencies measured, although a slight divergence in the plots might be seen beginning at about 13 GHz. If that is in-

deed a divergence, it may be caused by non-TEM propagation mechanisms, as noted in the Introduction.

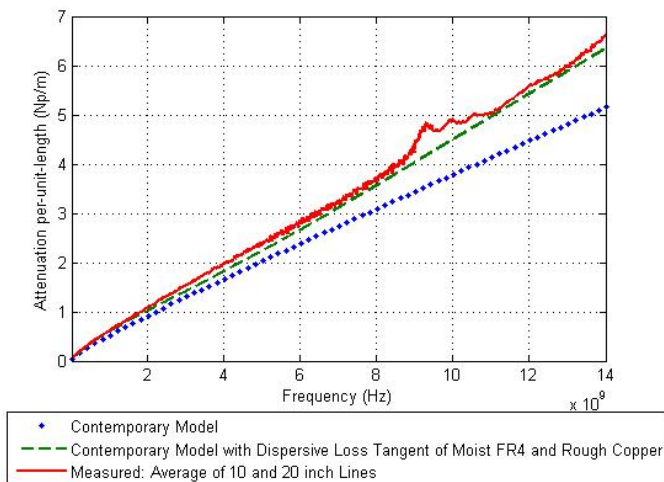


Figure 7: Final Contemporary Model with Dispersive Loss Tangent, Moisture and Surface Roughness Plotted with Benchmark Measured and Modeled Data

V. CONCLUSIONS

Current contemporary modeling techniques are available for predicting stripline losses, although it was found that their accuracy tends to become limited at frequencies above 2 GHz. Since there is growing demand for accurate stripline loss prediction for higher frequencies, the existing models may need to be modified to include other loss mechanisms.

This paper proposes an improved estimation technique for transmission-line attenuation using a modified contemporary model. Rather than depending on sometimes unreliable loss tangent specifications, the effective permittivity, determined by using a dielectric mixture model to approximately account for the effects of moisture, is applied to the Kramers-Kronig Relations to determine the frequency-dependent loss tangent. Further, the conduction losses in the contemporary model are modified to include the effect of surface roughness for the copper planes. The attenuation estimates of the modified model agree closely with the attenuation measurements for frequencies up to 14 GHz. This close agreement demonstrates that the additional loss mechanisms incorporated into the contemporary model were accounted for properly.

Of the parameters associated with transmission-line loss, the one that appears to have the greatest impact is moisture content. This is evident in the significant change in model results when going from a dry dielectric (Figure 5) to a dielectric containing realistic moisture (Figure 6). Certainly, loss does vary with other material parameters, but moisture appears to be dominant at the higher frequencies investigated in this study.

VI. ACKNOWLEDGMENTS

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